

Assignment 5

Note: There can be more than one correct answers.

Q1

[0.1 points]

Recall the following grammar:

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Give parse trees and derivations for each string.

- | | |
|----------|------------|
| a. a | c. $a+a+a$ |
| b. $a+a$ | d. $((a))$ |

A1

a)

Parse tree:

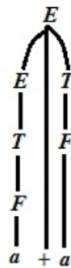


Derivation:

$$E \Rightarrow T \Rightarrow F \Rightarrow a$$

b)

Parse tree:

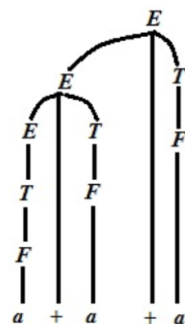


Derivation:

$$E \Rightarrow E + T \Rightarrow T + T \Rightarrow F + T \Rightarrow a + T \Rightarrow a + F \Rightarrow a + a$$

c)

Parse tree:

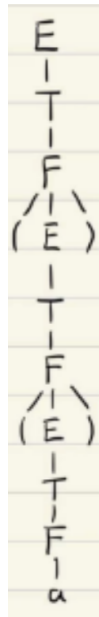


Derivation:

$$E \Rightarrow E + T \Rightarrow E + T + T \Rightarrow T + T + T \Rightarrow F + T + T \Rightarrow a + T + T \Rightarrow a + F + T \Rightarrow a + a + T \Rightarrow a + a + F \Rightarrow a + a + a$$

d)

Parse tree:



Derivation:

$$E \Rightarrow T \Rightarrow F \Rightarrow (E) \Rightarrow (T) \Rightarrow (F) \Rightarrow ((E)) \Rightarrow ((T)) \Rightarrow ((F)) \Rightarrow ((a))$$

Q2

Answer each part for the following context-free grammar G .

$$\begin{aligned} R &\rightarrow XRX \mid S \\ S &\rightarrow aTb \mid bTa \\ T &\rightarrow XTX \mid X \mid \epsilon \\ X &\rightarrow a \mid b \end{aligned}$$

- | | |
|--|--|
| a. What are the variables of G ? | i. True or False: $T \xRightarrow{*} T$. |
| b. What are the terminals of G ? | j. True or False: $XXX \xRightarrow{*} aba$. |
| c. Which is the start variable of G ? | k. True or False: $X \xRightarrow{*} aba$. |
| d. Give three strings in $L(G)$. | l. True or False: $T \xRightarrow{*} XXX$. |
| e. Give three strings <i>not</i> in $L(G)$. | m. True or False: $T \xRightarrow{*} XXXX$. |
| f. True or False: $T \Rightarrow aba$. | n. True or False: $S \xRightarrow{*} \epsilon$. |
| g. True or False: $T \xRightarrow{*} aba$. | o. Give a description in English of $L(G)$. |
| h. True or False: $T \Rightarrow T$. | |

A2

- (a) R, X, S, T ; (b) a, b ; (c) R ; (d) Three strings in $L(G)$ are ab, ba , and aab ;
 (e) Three strings not in $L(G)$ are a, b , and ϵ ; (f) False; (g) True; (h) False;
 (i) True; (j) True; (k) False; (l) True; (m) True; (n) False; (o) $L(G)$ consists of all strings over a and b that are not palindromes.

Note: (i) is true because the definition of $\xRightarrow{*}$ is:

If u, v , and w are strings of variables and terminals, and $A \rightarrow w$ is a rule of the grammar, we say that uAv **yields** uwv , written $uAv \Rightarrow uwv$. Say that u **derives** v , written $u \xRightarrow{*} v$, if $u = v$ or if a sequence $u_1, u_2, \dots, u_k$ exists for $k \geq 0$ and

$$u \Rightarrow u_1 \Rightarrow u_2 \Rightarrow \dots \Rightarrow u_k \Rightarrow v.$$

The **language of the grammar** is $\{w \in \Sigma^* \mid S \xRightarrow{*} w\}$.

According to the textbook, k can be 0.

Q3

[0.6 points]

Give context free grammars that generate the following languages. The alphabet is $\Sigma = \{0,1\}$.

- $\{w \mid w \text{ contains at least three 1s}\}$
- $\{w \mid w \text{ starts and ends with the same symbol}\}$
- $\{w \mid \text{the length of } w \text{ is odd}\}$
- $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- $\{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome}\}$
- The empty set

A3

a.

$$\begin{aligned} S &\rightarrow 0S \mid 1S_1 \\ S_1 &\rightarrow 0S_1 \mid 1S_2 \\ S_2 &\rightarrow 0S_2 \mid 1S_3 \\ S_3 &\rightarrow 0S_3 \mid 1S_3 \mid \varepsilon \end{aligned}$$

b.

$$\begin{aligned} S &\rightarrow 0S_0 \mid 1S_1 \\ S_0 &\rightarrow 0S_0 \mid 1S_0 \mid 0 \\ S_1 &\rightarrow 0S_1 \mid 1S_1 \mid 1 \end{aligned}$$

c.

$$\begin{aligned} S &\rightarrow 0S_o \mid 1S_o \\ S_o &\rightarrow 0S_e \mid 1S_e \mid \varepsilon \\ S_e &\rightarrow 0S_o \mid 1S_o \end{aligned}$$

d.

$$\begin{aligned} S &\rightarrow YSY \mid 0 \\ Y &\rightarrow 0 \mid 1 \end{aligned}$$

f.

e.

$$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \varepsilon$$

$$S \rightarrow S$$

Other solutions:

The context free grammar that generates the language

- (a) $\{\omega \mid \omega \text{ contains at least three 1s}\}$ is given by

$$\begin{aligned} S &\rightarrow P1P1P1P \\ P &\rightarrow 0P \mid 1P \mid \varepsilon \end{aligned}$$

The context free grammar that generates the language $\{\omega \mid \omega \text{ starts and ends with the same symbol}\}$ is given by

- (b) $\begin{aligned} S &\rightarrow 0P0 \mid 1P1 \mid 0 \mid 1 \\ P &\rightarrow 0P \mid 1P \mid \varepsilon \end{aligned}$

The context free grammar that generates the language

$\{\omega \mid \text{the length of } \omega \text{ is odd}\}$ is given by

- (c) $\begin{aligned} S &\rightarrow 0 \mid 1 \mid 00S \mid 01S \mid 10S \mid 11S \\ &\quad \text{(or)} \\ S &\rightarrow 0 \mid 1 \mid 0S0 \mid 0S1 \mid 1S0 \mid 1S1 \end{aligned}$

The context free grammar that generates the language $\{\omega \mid \text{the length of } \omega \text{ is odd and its middle symbol is a 0}\}$ is given by

- (d) $S \rightarrow 0 \mid 0S0 \mid 0S1 \mid 1S0 \mid 1S1$

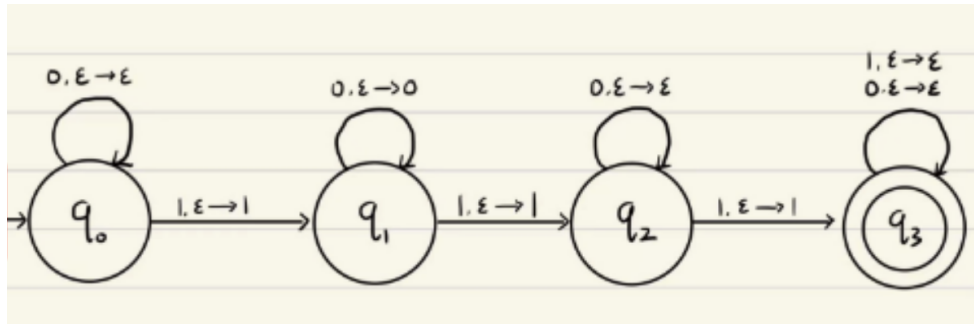
Q4

[0.5 points]

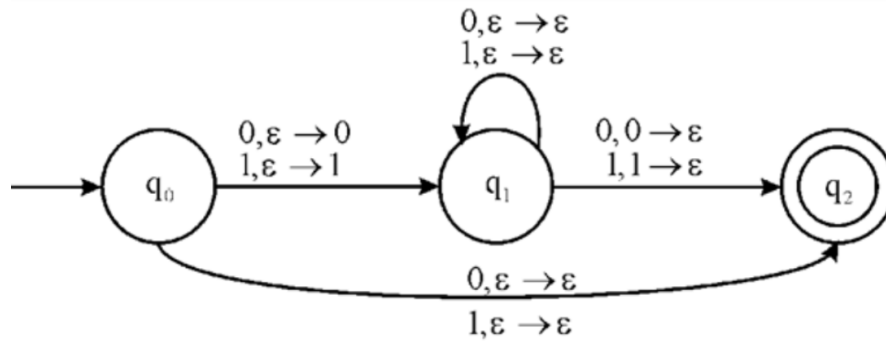
Give state diagrams of pushdown automata for the languages in exercise 3)

A4

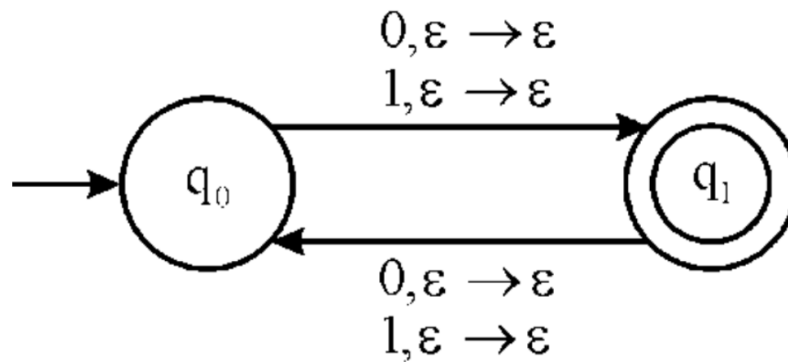
a) $\{w \mid w \text{ contains at least three 1s}\}$



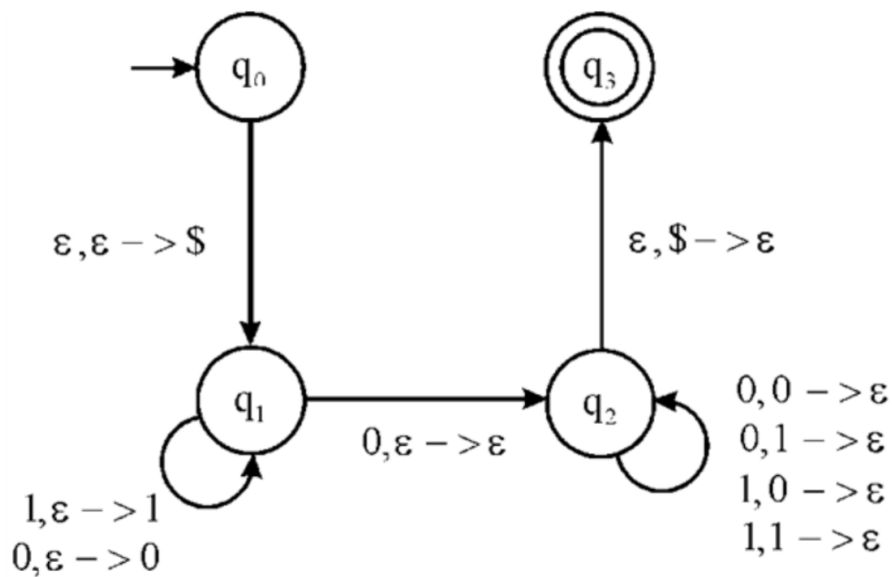
b) $\{w \mid w \text{ starts and ends with the same symbol}\}$



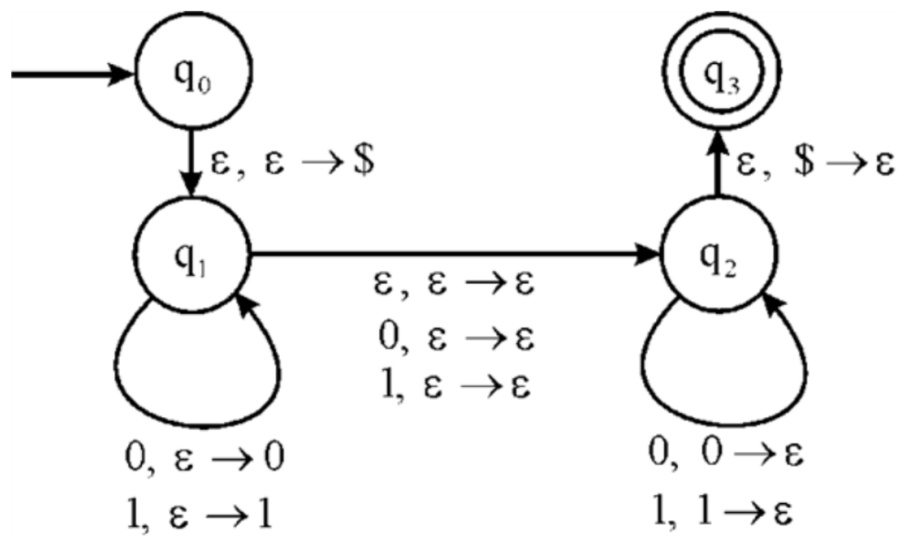
c) $\{w \mid \text{the length of } w \text{ is odd}\}$



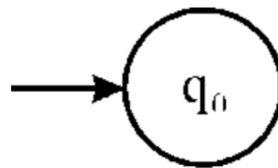
d) $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$



e) $\{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome}\}$



f) The empty set



Q5

[0.4 points]

Give context-free grammars generating the following languages.

- The set of strings over the alphabet $\{a,b\}$ with more a's than b's
- The complement of the language $\{a^n b^n \mid n \geq 0\}$

A5

(a)

(a) $S \rightarrow TaT$
 $T \rightarrow TT \mid aTb \mid bTa \mid a \mid \epsilon$
 T generates all strings with at least as many a's as b's, and S forces an extra a.

(b)

b. Suppose $A = \{a^n b^n \mid n \geq 0\}$, $B = \{a^m b^n \mid m \neq n\}$, $C = \{(a \cup b)^* ba(a \cup b)^*\}$ and $D = B \cup C$. Obviously, D is the complement of A.

First we give the CFG of B:

$$S \rightarrow aSb \mid A \mid B$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

Then we give the CFG of C:

$$S \rightarrow CbaC$$

$$C \rightarrow aC \mid bC \mid \varepsilon$$

So the CFG of D is:

$$S \rightarrow S_1 \mid S_2$$

$$S_1 \rightarrow aS_1b \mid A \mid B$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$S_2 \rightarrow CbaC$$

$$C \rightarrow aC \mid bC \mid \varepsilon$$

Q6

[0.25 points]

Convert the following CFG into an equivalent in Chomsky Normal Form using the procedure discussed at lecture.

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 11 \mid \varepsilon$$

A6

Step 1: add a new start variable.

$$S_0 \rightarrow A$$

$$A \rightarrow BAB \mid B \mid \varepsilon$$

$$B \rightarrow 11 \mid \varepsilon$$

Step 2: remove all rules that contain ε

$$S_0 \rightarrow A \mid \varepsilon$$

$$A \rightarrow BAB \mid BA \mid AB \mid A \mid B \mid BB$$

$$B \rightarrow 11$$

Step 3: remove unit rules

- remove $A \rightarrow A$

$$S_0 \rightarrow A \mid \varepsilon$$

$$A \rightarrow BAB \mid BA \mid AB \mid B \mid BB$$

$$B \rightarrow 11$$

- remove $A \rightarrow B$

$$S_0 \rightarrow A \mid \varepsilon$$

$$A \rightarrow BAB \mid BA \mid AB \mid BB \mid 11$$

$$B \rightarrow 11$$

- remove $S_0 \rightarrow A$

$$S_0 \rightarrow BA \mid AB \mid BAB \mid BB \mid 11 \mid \varepsilon$$

$$A \rightarrow BA \mid AB \mid BAB \mid BB \mid 11$$

$$B \rightarrow 11$$

Step 4: replace 1 by variable U

$$S_0 \rightarrow BA \mid AB \mid BAB \mid BB \mid UU \mid \varepsilon$$

$$A \rightarrow BA \mid AB \mid BAB \mid BB \mid UU$$

$$B \rightarrow UU$$

$$U \rightarrow 1$$

Step 5: break long rules

$$S_0 \rightarrow BA \mid AB \mid BA_1 \mid BB \mid UU \mid \varepsilon$$

$$A \rightarrow BA \mid AB \mid BA_1 \mid BB \mid UU$$

$$B \rightarrow UU$$

$$U \rightarrow 1$$

$$A_1 \rightarrow AB$$